

3 IoT-ENABLED DEVICE FOR SECURE DELIVERIES

With the outbreak of coronavirus pandemic, our reliance on doorstep delivery has surged. In these tough times, technology is playing a great role to ensure that deliveries reach us while we work safely from our homes. However, it is also important to ensure the safety of the products that need to be delivered. There are a lot of businesses that need confidential or expensive products delivered on a regular basis, such as jewellery or electronic devices. And it's important that these kinds of items reach safely in the hands of customers.

To solve these issues, Bengaluru-based startup Gordian Technologies has come up with the Internet of Things (IoT)-based security solutions for delivery companies. The company's IoT-enabled device called SecureIT makes use of a one-time password (OTP) to unlock itself, thereby ensuring security and safety.

This portable device can be unlocked only when the customer uses the OTP. It also provides real-time updates, GPS tracking, and internal cushioning for protection. With the use of such technology, losses due to theft, pilferage, and wrong handling of products can be completely eliminated.

The startup also provides a dash-

board to delivery businesses to keep a track of the products and get real-time updates on unlocks made at pick-up and drop. The company is currently targeting several verticals such as



electronics stores, jewellery delivery companies, CA and consultancy firms, banks, among others.

The SecureIT device is available on subscription and pay-per-use based models. The subscription model includes the dashboard, device maintenance, customer support, among others. Under the subscription model, the businesses rent out the device and get complete control over it. They can access the device, manage their deliveries, and get real-time updates.

Under the pay-per-use model, it offers secure deliveries as a service. Businesses can place the order on the startup's portal. A delivery executive will arrive for the pick-up and place the valuables in the device. Then,

the delivery executive will transfer the device to the customer, who will open it using the OTP received. Businesses will get complete information pertaining to the delivery, including timestamps and GPS locations of the device until it gets unlocked.

The startup aims to achieve 3,000 to 5,000 secured deliveries in a month on an average in the next three months. Following this, it will focus on raising a seed round for scaling up its operations across the country.

The startup currently faces stiff competition from some of the existing hyperlocal delivery companies. The current hyperlocal delivery companies provide low cost but less secured deliveries, while highly secured deliveries are generally expensive. However, the company is working towards providing a highly secured solution for deliveries at a lower cost. It also aims to have its own delivery fleet across the country.

In the times of Covid-19, when people can't go out of their houses, such technologies play an important role. Businesses and delivery companies can deliver their products on time without worrying about theft and losses, while customers can get their products sitting at home.

—Pallavi Singh

4 ENHANCED TECHNIQUES FOR REVIVING GEOTHERMAL SECTOR

While we see clean energies like solar and wind flourishing nowadays, their counterpart—geothermal energy—has been a limited resource due to reasons such as exploration hazards and insufficient good productivity. The process for obtaining geothermal energy entails drilling deep wells into hot areas to directly produce steam, which can be then used to drive tur-

bines for electricity generation. The hot areas are places with permeable rocks where underground water, when heated up naturally from heat generated under the earth's crust, moves towards the surface in liquid or vapour form.

This acts as a 24/7 reliable energy source but only in specific areas. In the past, researchers have tried pumping water through artificial wells, but the

permeability results have varied based on different conditions. Further, any system developed needs to be safe to avoid contamination of water or frequent earthquakes. This is where Fervo Energy, a US-based geothermal energy startup founded in 2017, came up with an economical and novel design that incorporates technologies such as horizontal drilling and advanced distributed fibre-optic